

Manage Your Finances with Maxima

This 22nd article in the series on mathematics through open source takes the reader into personal financial management. The author demonstrates how personal financial computations can be done easily by using Maxima.

In today's credit-oriented world, loans, EMIs, the principal, interest rates, savings, etc. are commonly used terms.

How well do we really understand these terms, or for that matter, are we able to accurately compute the figures related to such matters? Do we just accept what the other party has provided us? Or should one go into the details to first understand and then double-check the numbers being presented to us? This is where Maxima's finance package comes to your rescue; you can actually verify figures without much involvement in the computational details. This article is going to walk you through the details of this package.

Basic operations

To be able to use the finance package of Maxima, the first thing to do, after getting the Maxima prompt, is to load the finance package, using `load()`. Then go ahead to do the various computations. `days360()` is the simplest function to give the number of days between two dates, assuming there are 360 days per year and 30 days per month – this is the

norm when calculating interest rates.

```
$ maxima -q
(%i1) load(finance);
(%o1) /usr/share/maxima/5.24.0/share/contrib/finance/finance.
mac
(%i2) days360(2014, 1, 1, 2014, 10, 1);
(%o2) 270
(%i3) days360(2014, 1, 1, 2014, 12, 31);
(%o3) 360
(%i4) days360(2014, 1, 1, 2014, 3, 1);
(%o4) 60
(%i5) days360(2014, 1, 1, 2015, 1, 1);
(%o5) 360
(%i6) quit();
```

Note that `days360()` takes 'from' and 'to' dates, each as a triplet of the year, month and date.

One common computation that we often deal with is the final amount we would get after applying a particular rate of compound interest on a particular principal amount – just use `fv(<rate>, <principal>, <period>)` for computing the future value of the `<principal>`, at the compound interest `<rate>`, after a certain `<period>`. As an example, what would be the future value of investing ₹10,000 for 10 years at the compound interest rate of 15 per cent per year – just call `fv(0.15, 10000, 10)`;

